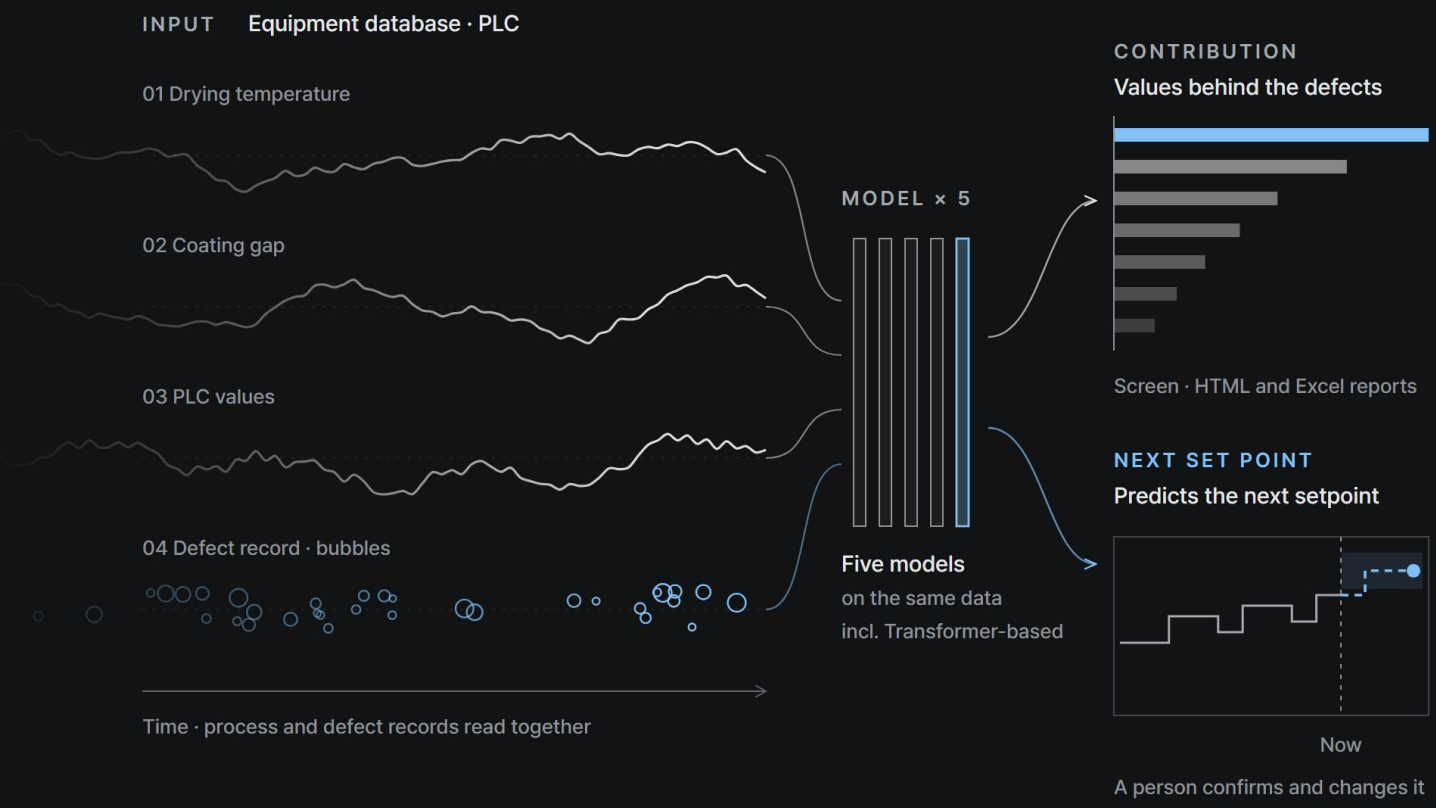
MACHINE LEARNING SOFTWARE FOR PROCESS DATA · PROCESS CONTROL

IntLearn ML

From the process data your equipment records, find the cause of defects and the next setpoints

Reads the process records, defect records and PLC values in the equipment database
Five models compared on the same data · how much each process value moved the defects · HTML and Excel reports



Process values and defect records go in; the values behind the defects and the next setpoint come out · concept diagram, example data

02 — OVERVIEW

Find the cause of defects in process data, and set the line

IntLearn ML reads the process records and defect records that build up in equipment databases together, finds which process values drive the defects and predicts the next setpoints. It shows which inputs drive the result and keeps training results as reports.

FIG. 01 PROCESS CONTROL · INTLEARN ML



A film coating line and a program screen that analyzes its process data · 3D rendering, an example

4 steps

Connect to apply

From data to setpoints

5 models

Same data, compared

Transformer-based included

HTML

HTML and Excel reports

Contribution and error

PLC

Equipment data

Process, defect, PLC values

READS

What your equipment records

Loads the process records and defect records in the equipment database, and the PLC values, together.

GIVES

Defect causes and next setpoints

Shows which process values drive the defects and predicts the next setpoints; reports keep the results.

From inspection to the cause, and on to autonomous control.

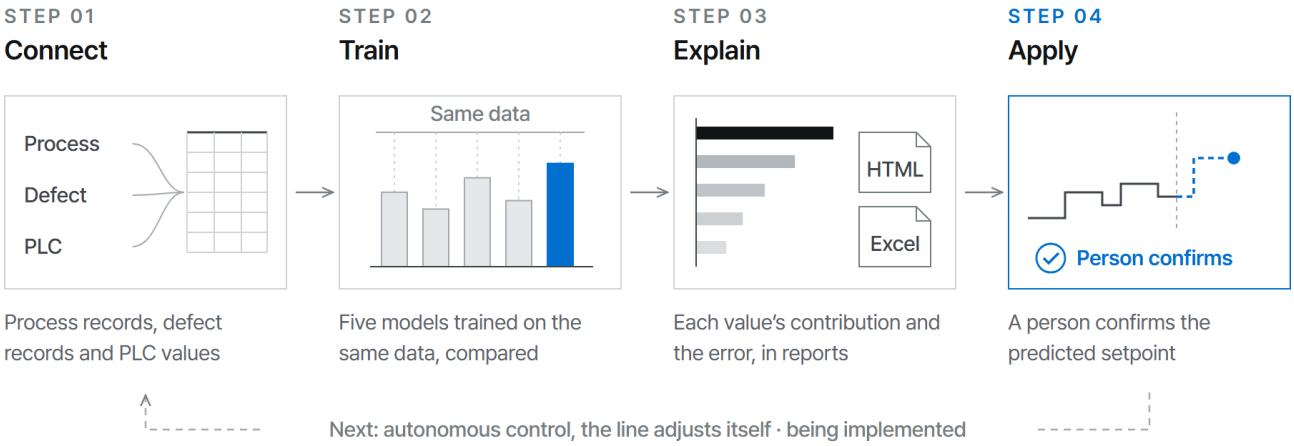
Beyond sorting out bad parts: it finds the process setpoints that made them.

03 — FLOW

From data to setpoints, in four steps

It loads the equipment's records, trains five models on the same data and compares them, shows which process value moved the result, and a person confirms the predicted setpoint before it is changed.

FIG. 02 CONNECT → TRAIN → EXPLAIN → APPLY



From data to setpoints

Product page 01 FLOW

01	Connect	Loads process records and defect records from the equipment database, and values from the PLC.
02	Train	Trains five model types, a Transformer-based one among them, on the same data and compares them.
03	Explain	Shows each input's contribution and error on screen and in HTML and Excel reports.
04	Apply	People confirm and apply the predicted setpoints, and the line moves on to autonomous control.

Which process value moved the defects, kept in a report.

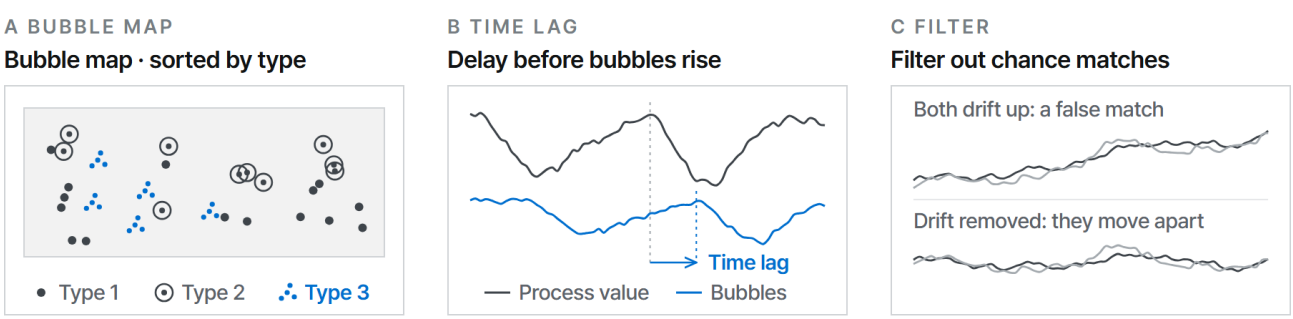
See each value's contribution and the prediction error on screen and in HTML and Excel reports, and keep the results.

04 — CAUSE ANALYSIS

Find which process value adds bubbles

Sorts the bubbles on the bubble map by type and checks how they follow the drying temperature and the coating gap, including how long bubbles take to rise after a change (time lag), to find which value adds bubbles. Values that only look linked because both drift the same way are filtered out.

FIG. 03 BUBBLE CORRELATION · EXAMPLE DATA



Finding which process value adds bubbles, in three steps · concept diagram, example data

FIG. 04 SOFTWARE · TIME SERIES



1 Pick process values 2 Trend line · moving average 3 Values ranked by contribution to a setpoint

IntLearn ML time-series analysis screen · example data (Korean UI)

Moving together does not make it the cause.

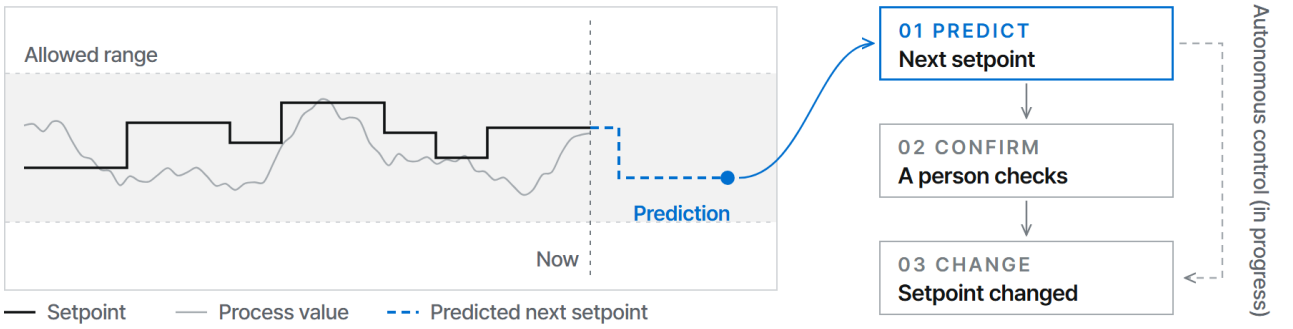
Chance matches are filtered out and the time lag is weighed, to find which value adds bubbles.

05 — SET POINT PREDICTION

Predict the next setpoint; a person confirms the change

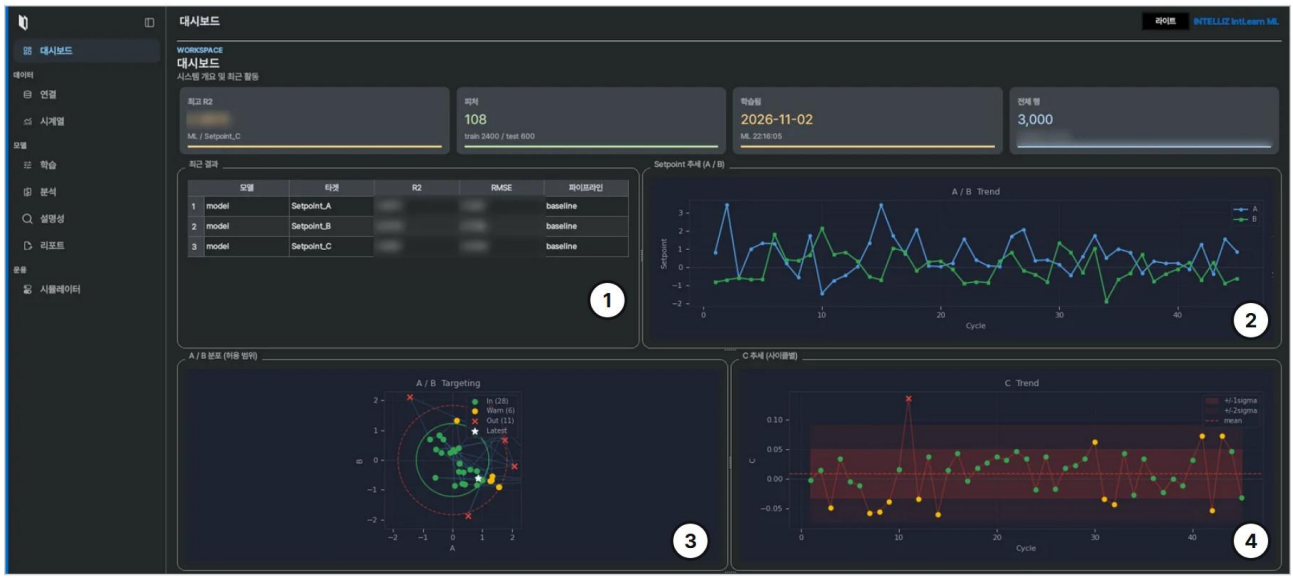
Predicts the next setpoints with a Transformer-based model, for example the drying temperature and the coating gap. Autonomous control, in which the line adjusts itself, is being implemented; today people confirm and change the setpoints.

FIG. 05 NEXT SET POINT · EXAMPLE DATA



The model predicts the next setpoint; a person confirms and changes it · concept diagram, example data

FIG. 06 SOFTWARE · DASHBOARD



1 Latest results per setpoint 2 Setpoint trends 3 Distribution within tolerance 4 Trend by cycle

IntLearn ML dashboard screen · example data (Korean UI)

The model predicts; a person decides.

Autonomous control is being implemented. Today a person confirms the setpoint IntLearn ML predicts, then changes it.

06 — CASE

Bubble correlation analysis and autonomous control for coated film

Finds the cause of bubbles on an adhesive coating film line from the drying temperature and the coating gap, and extends to autonomous control that predicts the next setpoints with a Transformer-based model.

FIG. 07 DEFECT CLINIC · EP.05



The film coating line in CEO Na's Defect Clinic EP.05 · 3D rendering, an example

CASE · PROCESS CONTROL

Process stage	Process Control
Inspection target	Continuous Material (Film/Paper/Coil)
Data	Bubbles · drying temperature · coating gap
Correction	Drying temperature · coating gap
Products	IntLearn ML
Industry	Consumer Goods, Plastic & Containers

A use case of IntLearn ML in paper and adhesive manufacturing. It uses the line's data to find how bubbles on the adhesive coating (laminating) line follow the drying temperature and the coating gap.

Autonomous control that predicts the next setpoints with a Transformer-based model is being implemented. Today people change the setpoints from the analysis.

The flow shown in Defect Clinic EP.05

Animated dramatization · autonomous control in progress

01 INSPECT

IntScan Pro logs bubbles

Replaces visual inspection: detects bubbles and logs their place on the film (MD/CD coordinates).

02 CAUSE

IntLearn ML finds why

Analyzes bubbles together with drying temperature and coating gap data to find the cause.

03 CONTROL

Setpoint correction

The line corrects its drying temperature and coating gap setpoints itself; being implemented in a Korean government project.

Use case page · QR

Bubble correlation analysis and autonomous control for coated film

intelliz.net/en/solutions/usecase

CEO Na's Defect Clinic EP.05 · 1:04 video

A Film Line That Fixes Its Own Bubbles?

intelliz.net/en/solutions/defect-clinic/ep-05

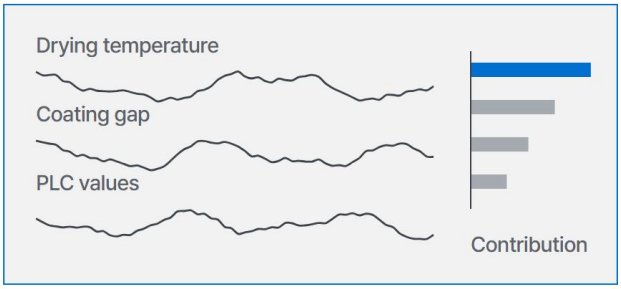
07 — LINEUP

Images go to IntLearn Pro, process data to IntLearn ML

In the INTELLIZ lineup, IntLearn ML covers process control. Next to the products that find defects in camera images, it finds the cause of defects and the next setpoints in the process data your equipment records.

FIG. 08 IMAGES · PROCESS DATA





IMAGES

IntLearn Pro

Deep learning visual inspection software

Reads	Inspection images taken by cameras
Gives	AI model and criteria in one file (.izpkg)

PROCESS DATA

IntLearn ML

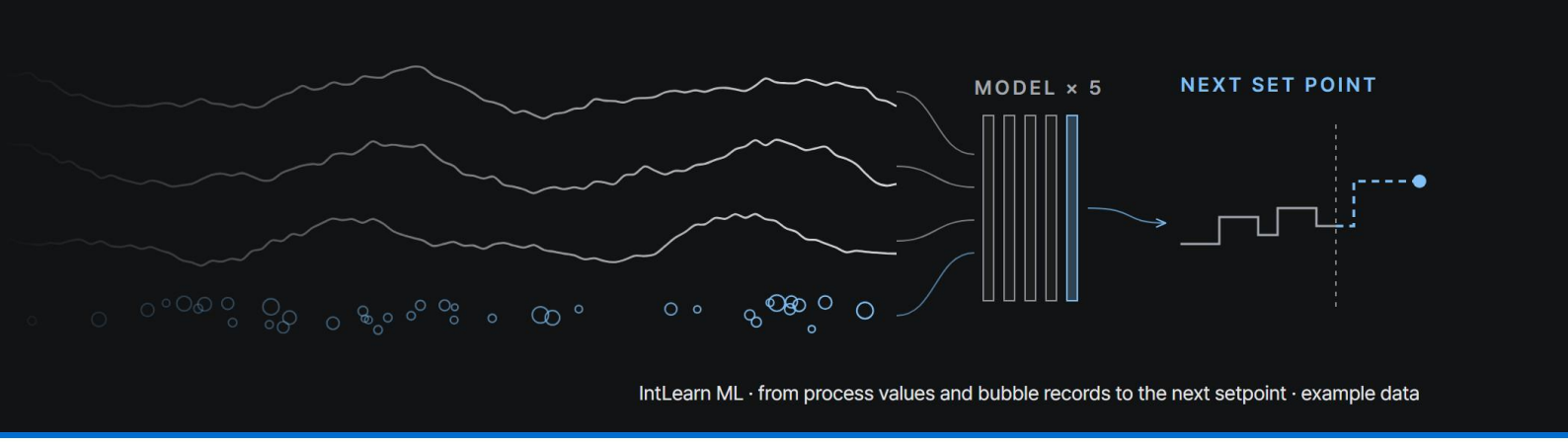
Machine learning that finds the cause of defects in process data and predicts setpoints

Reads	Process records · defect records · PLC values
Gives	Each value's contribution · reports · the next setpoint

Left: IntLearn Pro product image · right: process values and their contribution, an example drawing

Product lineup intelliz.net/en/products/lineup

01	Physical AI inspection platform	IntCloud Optic	AI optical inspection module that captures defects so they can be seen
		IntLearn Pro	Deep learning visual inspection software
		IntWise	Operate by speaking, and the know-how becomes a company asset
02	Physical AI platform	IntLearn Kinesis	Validate in a virtual factory before you install
03	Inspection runtime software	IntScan Pro	Runs line-scan camera inspection for continuous materials
		IntSight	Runs area-scan camera inspection for discrete parts
		IntViewer Pro	Monitors inspection results
04	Shop-floor data collection	IntCloud OCR	AI-OCR that reads shop-floor screens and gauges with a camera
05	Process control	IntLearn ML	Machine learning that finds the cause of defects in process data and predicts setpoints



IntLearn ML · from process values and bubble records to the next setpoint · example data

08 — CONTACT · NEXT STEPS

Find the cause in your process data

Tell us what your equipment records and which defect you want to reduce. We start with how to connect.

01	Data your equipment records: process, defects	02	Where the data builds up: database · PLC
03	The defect to reduce, and its record	04	Setpoints you change (e.g. drying temperature)

Next steps

Linked from the product and case pages · intelliz.net/en

Use case	Bubble correlation analysis and autonomous control for coated film intelliz.net/en/solutions/usecase/consumer-plastic/process-autonomous-control
CEO Na's Defect Clinic	A Film Line That Fixes Its Own Bubbles? · EP.05 intelliz.net/en/solutions/defect-clinic/ep-05
Guide	Adding AI to an Existing Inspection Machine: Checklist intelliz.net/en/guides/retrofit-ai-camera-plc-software
Company news	INTELLIZ's Three-Year National R&D Project Targets Autonomous Control of Drying Temperature and Coating Gap intelliz.net/en/company/news/film-coating-rnd
Product lineup	INTELLIZ product lineup · Process control intelliz.net/en/products/lineup

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Product page

See the use case

INTELLIZ

© INTELLIZ Corp. 2026 · Product information is based on public material at intelliz.net (2026-11-03).
The cover, back cover, charts and flow diagrams are concept drawings on example data, not measured results.
The program screens show IntLearn ML on example data (Korean UI); names are made up, scores blurred.
The coating line and the Defect Clinic EP.05 scene are 3D renderings from the INTELLIZ website.
CEO Na's Defect Clinic is an animated dramatization; IntLearn ML autonomous control is being implemented.